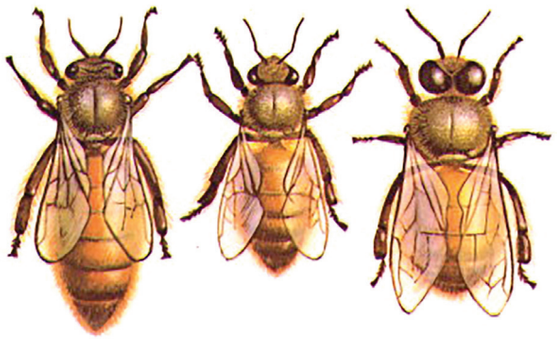


Queen Worker Drone

Figure 12.4 Social organization of honey bees

Then she becomes a soldier and guards the bee hive. In the second half her life lasting for three weeks, she searches and gathers the pollen, nectar, propolis and water.

The **drone** is the functional male member of the colony which develops from an unfertilized egg. It lives in a chamber called drone cell. Drones totally depend on workers for honey. The sole duty of the drone is to fertilize the virgin queen hence called "King of the colony". During swarming (the process of leaving the colony by the queen with a large group of worker bees to form a new colony) the drones follows the queen, copulates and dies after copulation.

Structure of a Bee Hive

The house of honey bee is termed as bee hive or comb. The hive consists of hexagonal cells made up of wax secreted by the abdomen of worker bees arranged in opposite rows on a common base. These hives are found hanging vertically from the rocks, building or branches of trees. The young stages of honey bees accommodate the lower and central cells of the hive called the **brood cells**. In *Apis dorsata*, the brood cells are of similar in size and shape but in other species, brood cells are of three types *viz*., queen cell for queens, worker cell for workers and drone cells for drones (Figure.12.5). The cells are intended for storage of honey and pollen in the upper portion of the comb whereas the lower portions are for brood rearing.

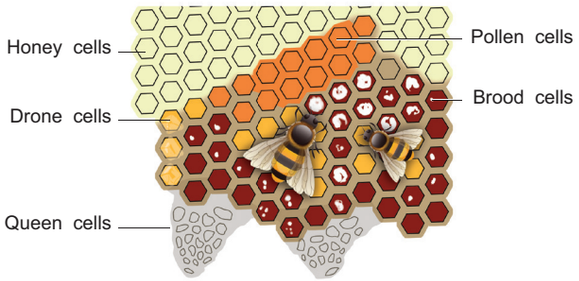


Figure 12.5 Structure of a hive showing various cells

Methods of Bee keeping

The main objective is to get more and more quality honey. There are two methods used by apiculturists. They are indigenous method and the modern method. In indigenous method, the honey extracted from the comb contains wax. To overcome the drawbacks of the indigenous method, the modern method has been developed to improve the texture of hives. In India, there are two types of beehives in practice namely, **Langstroth** and **Newton.**

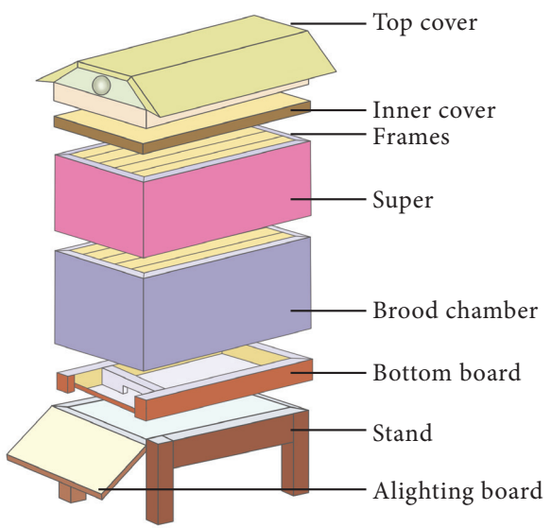


Figure 12.6 - Langstroth bee hive

TN_GOVT_XI_Bio Zoology_Ch12.indd 224 09-01-2020 5.33.17 PM